

Table 13: Prompt used to predict future trend (Part II)

1. Reflection: Reflect on scenarios where a specific method encounters significant challenges. Consider potential solutions that could effectively address these issues, make the solutions sounds reasonable, novel and amazing.

2. Analogy: Identify a specific problem you are currently facing and research existing solutions that have successfully tackled similar challenges. Explore these solutions and adapt key principles and strategies to your situation. Think creatively about how tools and approaches from other domains can be re-imagined to devise a novel strategy for your issue. Encourage you to actively explore methods in other fields to solve your current problems.

3. Deep Dive: Some methods may present specific approaches to addressing a particular problem. Consider whether there are aspects that could be modified to enhance their rationale and effectiveness.

Note: Each article's limitations are specific to that particular piece and should not be applied to others. Carefully consider the task at hand and analyze the potential issues you might encounter if you proceed with your original approach, reflecting on the challenges previously faced. Then, think critically about how to address these issues effectively.

You are encouraged to apply human reasoning strategies to identify future research directions based on prior studies. Aim for in-depth analysis rather than mere integration of existing ideas. Please avoid introducing unfamiliar information, ensuring that the trends you present are both authentic and reasonable. Before proposing any trends, take a moment to reflect on the principles underlying the methods you're employing and assess their relevance to your research area.

The future research direction should be related to the topic:
[Topic]

Please present the future research direction in the following format:

Future direction: ...

Table 14: Prompt used to generate idea (part I)

You are a scientific expert tasked with formulating a novel and innovative research idea based on your comprehensive literature review. Your objective is to propose a feasible approach that could significantly advance the field.

The following are examples of ideas you have proposed in the past that are similar to real papers. Please avoid this situation as much as possible. You can continue to make in-depth innovations, but avoid plagiarism: **[Bad case]**

Here are the entities you need to know: **[Entities]**

The topic you are studying is: **[Topic]**

The literature you have studied is as follows: **[Chain of ideas]**

Your idea is composed of the following components:

Motivation:

1. Provide a background for your idea, summarizing relevant work.
2. Identify shortcomings in previous research and highlight the specific problems that remain unsolved and that you aim to address.

Novelty:

1. Distinguish your proposed method from existing methods (preferably by naming specific approaches).
2. Detail the improvements of your method compared to past work.
3. Clearly outline at least three contributions your idea offers to the field, including the problems it resolves and the benefits it delivers.

Method:

1. Present a detailed description of your idea, focusing on the core method, the specific problem it solves, and enhancements over earlier research (citing relevant literature with titles).
2. Explain the step-by-step methodology, including the functions of each module and the rationale for why this approach effectively addresses previous challenges.

Please adhere to the following guidelines:

1. Your research idea should be innovative, feasible, and contribute meaningfully to the field. Please carefully examine the idea you have proposed, avoid immediate perception, and try to be different from the previous methods as much as possible.
2. Ensure your proposal is solid, clearly defined, and practical to implement. Logic should underpin your reasoning.
3. Write in clear, concise language aimed at an audience with limited background knowledge in the subject. Avoid complex technical jargon, but when professional terms are necessary, provide thorough explanations.
4. Refrain from introducing concepts from uncertain fields to prevent proposing ideas that may be incorrect or impractical.
5. When referencing other research, please include the titles of the cited papers.
6. Please avoid introducing unfamiliar information, ensuring that the trends you present are both authentic and reasonable. Before proposing any trends, take a moment to reflect on the principles underlying the methods you're employing and assess their relevance to your research area.

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Table 15: Prompt used to generate idea (part II)

7. Each article’s limitations are specific to that particular piece and should not be applied to others. Carefully consider the task at hand and analyze the potential issues you might encounter if you proceed with your original approach, reflecting on the challenges previously faced. Then, think critically about how to address these issues effectively.

The following section delineates the progressive relationships among the previously summarized research papers: **[Trend]**

The following section outlines the potential future research directions based on the literature you have studied: **[Future direction]**

Please output your motivation, novelty, method firstly and then output your final idea. The final idea should clearly explain the origins, motivation, and challenges of your idea, detailing how you overcame these hurdles.

Please present the final idea in the following format:

Motivation: ...

Novelty: ...

Method: ...

Final idea: ...

Table 16: Prompt used to check the novelty of the idea

You are a scientific research expert tasked with evaluating the similarity between a specified idea and existing research. Your objective is to determine if the target idea closely resembles any findings in the provided papers.

The target idea you need to check is as follows: **[Idea]**

The relevant papers you need to refer to are as follows: **[Content of retrieved papers]**

Here are your guidelines:

1. Comparison Process: Begin by thoroughly comparing each paper’s ideas with the target idea. Consider the methodologies, conclusions, and underlying concepts in each paper in your analysis.
2. Similarity Assessment: If the target idea shares fundamental similarities with any existing research to the extent that they can be considered identical, classify this as plagiarism.
3. Output: Your output should provide a clear thought process, the similarity assessment, a summary of the target idea, and the ID of the most relevant similar paper.

Please output strictly in the following format:

Think: ...

Similar: 0/1

Summary of the idea: ...

Similar paper id: 0 to n